

Project Proposal

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A Big Data–Driven Customer Churn Prediction System Using Machine Learning

1. Problem Statement

Customer churn is a major challenge for many industries such as telecommunications, banking, and subscription-based services. Churn occurs when customers stop using a company's services and switch to competitors. Retaining existing customers is more cost-effective than acquiring new ones, which makes churn prediction an important business problem.

The objective of this project is to build an end-to-end system that uses machine learning and database technologies to predict customer churn. The project will store customer data in a relational database and apply classification algorithms to identify customers who are at risk of leaving. The system will also analyse the key factors that influence churn and present meaningful insights for decision-making.

This project solves a real-world problem and helps organisations improve customer retention strategies by using data-driven methods.

2. Dataset

The project will use a publicly available customer churn dataset obtained from Kaggle. The dataset contains approximately 10,000 to 100,000 customer records and includes attributes such as:

- **Customer demographics:** gender, senior citizen status, partner, and dependents
- **Service information:** phone service, multiple lines, internet service type (DSL, fiber optic, or none), and additional services such as online security, device protection, and streaming services
- **Account details:** tenure, contract type, payment method, and paperless billing status
- **Billing information:** monthly charges and total charges
- **Target variable:** Churn (Yes/No), indicating whether a customer has discontinued the service

The dataset will be stored in a relational database such as MySQL. SQL queries will be used for data exploration, filtering, and preprocessing before applying machine learning models.

3. Anticipated Methods

The project will follow a step-by-step data analysis and machine learning pipeline:

a) Data Preprocessing

- Import the dataset into a relational database

- Handle missing values and remove inconsistencies
- Convert categorical variables into numerical form using encoding techniques
- Normalize numerical features
- Create new features using SQL queries

b) Machine Learning Models

Several classification algorithms will be implemented and compared, including:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting (XGBoost or LightGBM)

Models will be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The best model will be selected for further analysis.

c) Model Interpretation

Explainable AI techniques will help interpret predictions:

- **SHAP** to identify important features
- **LIME** to explain individual predictions

These tools will answer questions like: Which factors influence churn the most? Why is a specific customer predicted to churn?

4. Related Literature with references:

Previous research shows that ML methods are effective for predicting customer churn. Verbeke et al. (2012) used logistic regression and decision tree models to predict churn in the telecom industry. Later research found that ensemble models such as Random Forest and Gradient Boosting provide better accuracy (Idris et al., 2019).

Explainable AI has become important in recent years. LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017) help explain how machine learning models make predictions and increase trust in their results.

This project combines databases, machine learning, and explainable AI in a practical churn prediction system.

5. Expected Outcomes

- A functional churn prediction system using real-world data
- Comparison of different machine learning models
- Identification of key features affecting churn
- Explainable predictions using SHAP and LIME
- A final report and presentation demonstrating technical and business insights